Guangdong Technion – Israel Institute of Technology (GTIIT)

125101 – ANALYTICAL CHEMISTRY 1 FOR ENGINEERS

FINAL EXAMINATION A

2025 SPRING TERM

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24, June 2025 TIME ALLOWED: **3 HOURS**

Name: ID:

INSTRUCTIONS TO STUDENTS

1. The final exam includes 16 multiple choice questions (MCQ), 6 of them worth 5 points, 10 of them worth 7 points.
2. Circle a single answer on the exam papers and mark it in the answer sheet properly. Marking of several answers will be considered as if no answer was selected at all.
4. Students can use their own calculators and hard copy dictionaries.
5. Necessary equations are provided in the **Appendix** at the end.

Appendix

Some formulas are following, which may be used in the exam.

(1) Standard addition method:

A solution with successive additions $I_{(X+S)}(\frac{V}{V_o})=I_X+\frac{I_X}{[X]_i}[S]_i(\frac{V_s}{V_o})$

Multiple solutions with same final volume $I_{(X+S)}=\frac{I_X}{[X]_i}[X]_f+\frac{I_X}{[X]_i}[S]_f$

(2) Henderson Equation: $pH = pKa + \log\frac{[A^-]}{[HA]}$

(3) Buffer capacity: $\beta = 2.3C\times\alpha(HA)\times\alpha(A^-) \Rightarrow \beta_{max.} = 0.575\ C$

(4) Standard deviation of random variables: $s=\sqrt{\frac{\sum_i(x_i-\bar{x})^2}{N-1}}$

(5) Method of least squares for y = *k*x + c:

$$k=\frac{N\left(\sum x_i y_i\right)-\left(\sum x_i\right)\left(\sum y_i\right)}{N\left(\sum x_i^2\right)-\left(\sum x_i\right)^2} \qquad c=\frac{\left(\sum x_i^2\right)\left(\sum y_i\right)-\left(\sum x_i\right)\left(\sum x_i y_i\right)}{N\left(\sum x_i^2\right)-\left(\sum x_i\right)^2}$$

(6) Titration error in Acid-base titration:

$$\text{Titration error}=\frac{C+C_0}{CC_0}\left[\frac{Kw}{[H^+]_{ep}}-[H^+]_{ep}\right]-\frac{[H^+]_{ep}}{Ka} \quad \text{Weak acid}$$

$$\text{Titration error}\approx\frac{C+C_0}{CC_0}\left[\frac{Kw}{[H^+]_{ep}}-[H^+]_{ep}\right] \quad \text{Strong acid}$$

(7) Gran plot equations:

For acid-base titration, $V_b\times10^{-pH} = K_aV_e - K_aV_b$

For redox titration, $V\times10^{-E/0.05916} = Ve\times10^{(Eref - Eo)/0.05916} - V\times10^{(Eref - Eo)/0.05916}$

(8) Nernst equation: aA + bB ⇌ cC + dD

$$E=E^\circ-\frac{RT}{nF}\ln\frac{[C]^c[D]^d}{[A]^a[B]^b}=E^\circ-\frac{0.059\ 16}{n}\log\frac{[C]^c[D]^d}{[A]^a[B]^b}$$

(9) Coulometric analysis: $Q = n_e\times F = I_A \times t_{sec}$ ($F = 9.6485\times10^4$ C/mol; A=1C/s)

(10) Contaminants in the precipitate's liquid after *n* washing cycles,

$$X_n=X_0\left(\frac{u}{u+V/n}\right)^n$$